RF Components: Characterization and Measurements

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1 Aim

The aim of this experiment is to understand the operation and experimentally characterize different RF components used in an RF receiver chain which includes a filter, mixer, power-splitter, power-combiner, and RF amplifier.

2 Apparatus Required

RF signal generators, Spectrum analyzer, Digital storage oscilloscope, RF filter, RF mixer, Two-way power splitter/combiner, RF amplifier with DC power supply, RF Tee connector, Open-ended coaxial stub, 50 Ω terminations, RF coaxial cables and adapters.

3 Theory

3.1 RF Components in a Receiver Chain

An RF receiver processes radio-frequency signals using different components, each performing a specific function. A filter selects the required frequency range, a mixer changes the frequency of a signal, an amplifier increases signal power, and a splitter/combiner divides or combines RF signals. Characterizing these components experimentally allows their actual transmission, loss, gain, and nonlinear behaviour to be determined.

3.2 RF Filter

An RF filter is used to allow selected frequency components of a signal to pass while attenuating unwanted frequencies. The commonly used filter types are low-pass, high-pass, band-pass, and band-reject(notch) filters. A low-pass filter allows frequencies below its cutoff frequency to pass with relatively small attenuation, while higher frequencies are attenuated. The transmission performance of a filter is characterized by its insertion loss, defined as

$$IL = P_{\text{in}} - P_{\text{out}}, \tag{1}$$

where P_{in} is the power measured without the filter and P_{out} is the power measured after inserting the filter. The $-3\,\text{dB}$ cutoff frequency is determined relative to the minimum insertion loss in the passband.

3.3 Reflection Coefficient and Return Loss

When the input impedance of an RF component does not match the characteristic impedance of the transmission line, a portion of the incident signal is reflected. The voltage reflection coefficient is defined as

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \tag{2}$$

where Z_{in} is the input impedance of the component and Z_0 is the characteristic impedance of the transmission line. From the measured maximum and minimum voltages, the magnitude of the reflection coefficient is

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{3}$$

The return loss is given by

$$RL = -20 \log_{10} |\Gamma|. \tag{4}$$

A smaller reflection coefficient and larger return loss indicate better impedance matching.

3.4 Quarter-Wave Open-Circuit Stub

An open-ended coaxial stub can be used as a notch filter. When its physical length is approximately one-quarter of the guided wavelength,

$$L = \frac{\lambda}{4}, \tag{5}$$

the open circuit at the far end of the stub is transformed into a very low impedance at the Tee junction at the resonant frequency. As a result, the signal at this frequency is strongly attenuated, producing a notch in the transmission response. The expected notch frequency is

$$f_0 = \frac{V_F c}{4L}, \tag{6}$$

where V_F is the velocity factor of the cable, c is the speed of light in vacuum, and L is the physical length of the stub.

3.5 RF Mixer

An RF mixer is a three-port nonlinear device having RF, local oscillator (LO), and intermediate-frequency (IF) ports. It is used to translate a signal from one frequency to another. For an RF input frequency f_{RF} and a local oscillator frequency f_{LO}, the mixer produces sum and difference frequency components:

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} + f_{\mathrm{LO}}, \tag{7}$$

and

$$f_{\mathrm{IF}} = \left| f_{\mathrm{RF}} - f_{\mathrm{LO}} \right|. \tag{8}$$

The conversion loss is defined as

$$CL = P_{\mathrm{RF}} - P_{\mathrm{IF}}, \tag{9}$$

where P_{RF} is the RF input power and P_{IF} is the measured power of the selected IF component.

3.6 Power Splitter and Combiner

A power splitter divides an input RF signal into two or more output signals. For an ideal two-way equal power splitter, each output has a power level 3 dB below the input power.

The losses at the two output ports are given by

$$L_1 = P_{\mathrm{in}} - P_{\mathrm{out1}}, \tag{10}$$

$$L_2 = P_{\mathrm{in}} - P_{\mathrm{out2}}. \tag{11}$$

A power combiner performs the reverse operation by combining signals applied at its input ports. The output depends on the frequencies, amplitudes, and relative phases of the input signals.

3.7 RF Amplifier

An RF amplifier is used to increase the power of an RF signal. Under small-signal conditions, the amplifier gain is given by

$$G = P_{\mathrm{out}} - P_{\mathrm{in}}. \tag{12}$$

At sufficiently high input powers, the amplifier becomes nonlinear. This nonlinearity produces harmonics and causes the gain to decrease from its small-signal value.

The 1 dB compression point is defined as the input power at which the amplifier gain has decreased by 1 dB from its small-signal gain.

When two input tones at frequencies f_1 and f_2 are applied to a nonlinear amplifier, third-order intermodulation products are generated at

$$2f_1 - f_2 \tag{13}$$

and

$$2f_2 - f_1. \tag{14}$$

The fundamental output increases approximately with a $1:1$ slope, while the third-order intermodulation products increase with a $3:1$ slope on a logarithmic scale. The extrapolated intersection of these two characteristics is called the third-order intercept point (TOI).

4 Experimental Procedure

4.1 Filter Characterization

4.1.1 A1. Transmission Response and Insertion Loss

The RF signal generator was connected to the spectrum analyzer through a reference cable, and the input power was measured at selected frequencies. The filter was then inserted without changing the generator settings or connections. The output power was recorded at the same frequencies and the insertion loss was calculated.

4.1.2 A2. Reflection Coefficient and Return Loss

The RF generator, Tee connector, oscilloscope, and filter were connected as shown in the reflection measurement setup. The maximum and minimum voltage amplitudes were measured over a small frequency range in the passband. These values were used to calculate the reflection coefficient and return loss.

4.1.3 A3. Quarter-Wave Open-Circuit Stub

The open-ended coaxial stub was connected to the Tee junction between the RF generator and spectrum analyzer. The expected notch frequency was calculated from the stub length and velocity factor. The output power was measured over the selected frequency range with the stub connected, and the frequency of maximum attenuation was identified.

4.2 Mixer Characterization

The RF and LO ports of the mixer were connected to separate signal generators, while the IF port was connected to the spectrum analyzer. The expected sum and difference frequencies were calculated. The corresponding IF output powers were measured and used to determine the conversion loss.

4.3 Power Splitter/Combiner Characterization

4.3.1 C1. Power Splitter

The input of the two-way splitter was connected to the RF generator. One output was connected to the spectrum analyzer while the other was terminated with a $50\,\Omega$ load. The output powers from both ports were measured and the corresponding losses were calculated.

4.3.2 C2. Power Combiner

Two signal generators were connected to the input ports of the power combiner and the combined output was connected to the spectrum analyzer. The output frequency components and power levels were measured.

4.4 Amplifier Characterization

4.4.1 D1. Gain versus Frequency

The amplifier was connected to the specified DC supply and operated at a fixed input power. The output power was measured at different frequencies and the gain was calculated.

4.4.2 D2. 1-dB Compression Point

The amplifier input power was gradually increased from a low level while recording the output power. The gain was calculated at each input power, and the 1-dB compression point was determined from the reduction in gain relative to the small-signal value.

4.4.3 D3. Third-Order Intercept Point

Two RF tones were applied to the amplifier and the input power per tone was varied. The fundamental output and third-order intermodulation product powers were recorded. The TOI was determined by extrapolating the fundamental and third-order characteristics.

5 Observations and Calculations

5.1 A1. Filter Transmission Response

The measured reference input power was

$$P_{\text{in}} = -0.90\,\text{dBm}.$$

The insertion loss was calculated using

$$IL = P_{\text{in}} - P_{\text{out}}. \tag{15}$$

Table 1: Measured filter transmission response.

Frequency (MHz)	P_{out} (dBm)	IL (dB)
100	-1.36	0.46
150	-1.59	0.69
200	-1.84	0.94
250	-1.70	0.80
300	-1.89	0.99
350	-2.50	1.60
400	-2.22	1.32
450	-3.09	2.19
500	-3.26	2.36
550	-4.72	3.82
600	-8.79	7.89
650	-21.92	21.02
700	-28.95	28.05

The average insertion loss over the 100–300 MHz passband is

$$IL_{\text{pass}} = \frac{0.46 + 0.69 + 0.94 + 0.80 + 0.99}{5} \approx 0.78\,\text{dB}.$$

Hence, the -3 dB cutoff level is

$$IL_c = 0.78 + 3 = 3.78\,\text{dB}.$$

Interpolating between 500 MHz and 550 MHz gives

$$\boxed{f_c \approx 548\,\text{MHz}}.$$

Similarly, taking 20 dB insertion loss as the stop-band level gives

$$\boxed{f_s \approx 646\,\text{MHz}}.$$

The increasing insertion loss with frequency indicates a low-pass response.

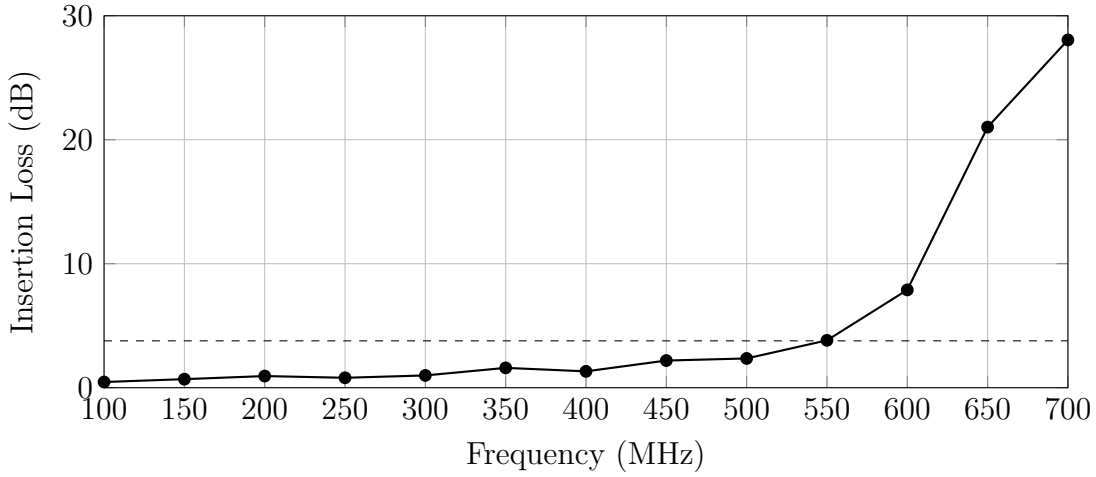


Figure 1: Measured insertion loss of the filter. The dashed line indicates the −3 dB cutoff level.

5.2 A2. Filter Reflection Coefficient and Return Loss

The maximum and minimum voltage amplitudes measured at the Tee junction were

$$V_{\max} = 136\,\text{mV}, \qquad V_{\min} = 132\,\text{mV}.$$

The reflection coefficient was calculated using

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{16}$$

Therefore,

$$|\Gamma| = \frac{136 - 132}{136 + 132} \approx 0.0149.$$

The return loss is

$$RL = -20 \log_{10} |\Gamma|. \tag{17}$$

Hence,

$$RL = -20 \log_{10}(0.0149) \approx 36.5\,\text{dB}.$$

The corresponding VSWR is

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} \approx 1.03.$$

Thus, the measured reflection coefficient, return loss, and VSWR are approximately 0.0149, 36.5 dB, and 1.03, respectively.

5.3 A3. Quarter-Wave Open-Circuit Stub

The measured stub length was

$$L = 0.91\,\mathrm{m},$$

with velocity factor

$$V_F = 0.66.$$

The expected notch frequency was calculated from

$$f_0 = \frac{V_F c}{4L}. \tag{18}$$

Using $c = 3 \times 10^8\,\mathrm{m/s}$,

$$f_{0,\mathrm{calc}} = \frac{0.66(3 \times 10^8)}{4(0.91)} \approx 54.40\,\mathrm{MHz}.$$

The input power was maintained at

$$P_{\mathrm{in}} = -0.90\,\mathrm{dBm}.$$

The attenuation was calculated using

$$A = P_{\mathrm{in}} - P_{\mathrm{out}}. \tag{19}$$

Table 2: Measured response of the quarter-wave open-circuit stub.

Frequency (MHz)	P_{out} (dBm)	Attenuation (dB)
50	-9.20	8.30
51	-10.30	9.40
52	-11.40	10.50
53	-13.20	12.30
54	-15.40	14.50
55	-29.40	28.50
56	-22.60	21.70
57	-27.80	26.90
58	-25.40	24.50
59	-20.60	19.70
60	-17.50	16.60
61	-15.10	14.20

The maximum attenuation was observed at

$$\boxed{f_{0,\mathrm{meas}} = 55\,\mathrm{MHz}},$$

with

$$\boxed{A_{\max} = 28.5\,\mathrm{dB}}.$$

The difference between the calculated and measured notch frequencies is

$$\Delta f = 55 - 54.40 = 0.60\,\mathrm{MHz},$$

corresponding to approximately 1.1% difference.

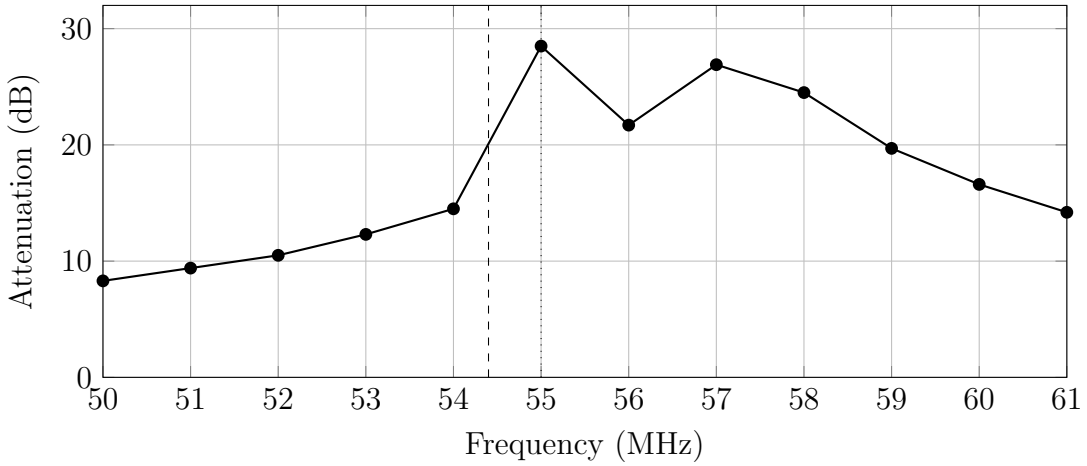


Figure 2: Measured attenuation of the quarter-wave open-circuit stub. The dashed and dotted lines indicate the calculated and measured notch frequencies, respectively.

5.4 B. Mixer Characterization

The LO frequency was maintained at

$$f_{\mathrm{LO}} = 10\,\mathrm{MHz},$$

with LO power

$$P_{\mathrm{LO}} = 1.04\,\mathrm{dBm}.$$

The RF input power was maintained at

$$P_{\mathrm{RF}} = -0.90\,\mathrm{dBm}.$$

For each RF frequency, the expected sum and difference frequencies were calculated using

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} + f_{\mathrm{LO}} \tag{20}$$

and

$$f_{\mathrm{IF}} = |f_{\mathrm{RF}} - f_{\mathrm{LO}}| \,. \tag{21}$$

The conversion loss was calculated from

$$L_{\mathrm{conv}} = P_{\mathrm{RF}} - P_{\mathrm{IF}}. \tag{22}$$

Table 3: Mixer characterization measurements.

f_{RF} (MHz)	Type	f_{IF} (MHz)	P_{IF} (dBm)	L_{conv} (dB)
13	Difference	3	-15.00	14.10
13	Sum	23	-13.80	12.90
23	Difference	13	-13.80	12.90
23	Sum	33	-14.10	13.20
33	Difference	23	-10.40	9.50
33	Sum	43	-10.08	9.18
43	Difference	33	-10.61	9.71
43	Sum	53	-10.57	9.67

The average conversion loss obtained from the measurements was

$$\boxed{L_{\mathrm{conv}} \approx 11.4\,\mathrm{dB}}.$$

The measured output frequencies correspond to the expected sum and difference frequency components generated by the mixer.

5.5 C1. Power Splitter

The input power was maintained at

$$P_{\mathrm{in}} = -0.90\,\mathrm{dBm}.$$

For an ideal two-way splitter, each output should be approximately 3 dB below the input. The losses were calculated using

$$L_1 = P_{\mathrm{in}} - P_{\mathrm{out1}}, \qquad L_2 = P_{\mathrm{in}} - P_{\mathrm{out2}}. \tag{23}$$

Table 4: Measured power division of the two-way splitter.

Frequency (MHz)	P_{out1} (dBm)	P_{out2} (dBm)	L_1 (dB)	L_2 (dB)
33	-4.42	-4.40	3.52	3.50
43	-4.46	-4.44	3.56	3.54
53	-4.47	-4.43	3.57	3.53

The average losses were

$$L_{1,\text{avg}} \approx 3.55\,\text{dB}, \qquad L_{2,\text{avg}} \approx 3.52\,\text{dB}.$$

Thus, the two output ports were approximately equally balanced, with about 0.5 dB additional loss compared with the ideal 3 dB division.

5.6 C2. Power Combiner

Two equal-frequency signals were applied to the two input ports:

$$f_1 = f_2 = 10\,\text{MHz},$$

with input powers

$$P_1 = -0.90\,\text{dBm}, \qquad P_2 = 1.64\,\text{dBm}.$$

The measured peak voltages were

$$V_1 = 201.8\,\text{mV}, \qquad V_2 = 270.4\,\text{mV}.$$

For coherent, in-phase signals,

$$V_{\text{sum}} = V_1 + V_2. \tag{24}$$

Therefore,

$$V_{\text{sum}} = 201.8 + 270.4 = 472.2\,\text{mV}.$$

For a 50 Ω system, the ideal combined power is

$$P_{\text{ideal}} = \frac{V_{\text{sum}}^2}{2R}. \tag{25}$$

Thus,

$$P_{\text{ideal}} = \frac{(0.4722)^2}{2(50)} \approx 2.23\,\text{mW} \approx 3.48\,\text{dBm}.$$

The measured combined output power was

$$P_{\text{meas}} = 3.05\,\text{dBm}.$$

Hence, the difference from the ideal value is

$$\Delta P = 3.48 - 3.05 = 0.43\,\text{dB}.$$

Thus, the measured combined power was approximately 0.43 dB below the ideal coherent value.

5.7 D1. Amplifier Gain versus Frequency

The amplifier input power was maintained at

$$P_{\mathrm{in}} = -20\,\mathrm{dBm}.$$

The output power was measured over the frequency range 100–200 MHz. The gain was calculated using

$$G = P_{\mathrm{out}} - P_{\mathrm{in}}. \tag{26}$$

Table 5: Measured amplifier gain versus frequency.

Frequency (MHz)	P_{out} (dBm)	Gain (dB)
100	2.68	22.68
110	2.65	22.65
120	2.58	22.58
130	2.52	22.52
140	2.43	22.43
150	2.34	22.34
160	2.30	22.30
170	2.27	22.27
180	2.24	22.24
190	2.23	22.23
200	2.22	22.22

The amplifier gain decreases gradually from

$$G(100\,\mathrm{MHz}) = 22.68\,\mathrm{dB}$$

to

$$G(200\,\mathrm{MHz}) = 22.22\,\mathrm{dB}.$$

Thus, the total gain variation over the measured frequency range is approximately 0.46 dB.

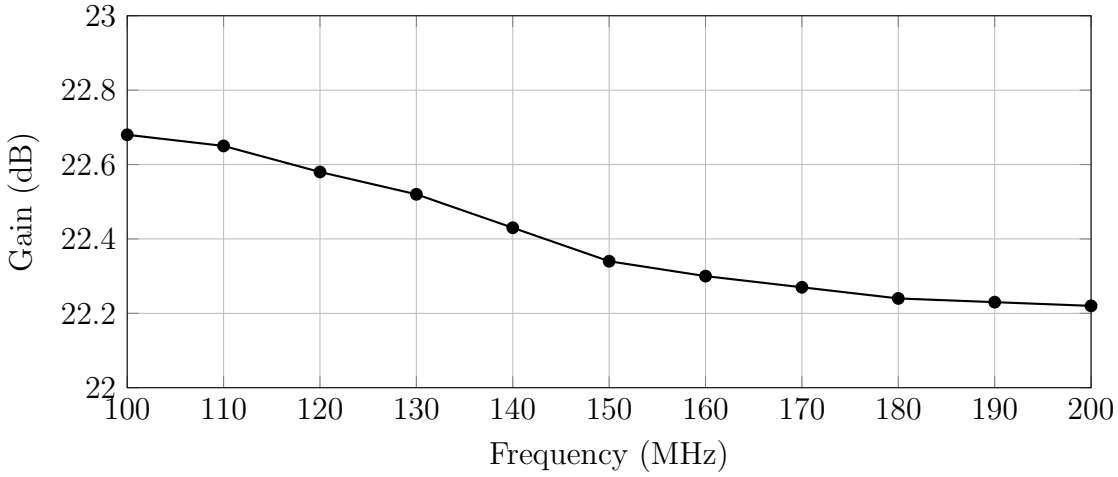


Figure 3: Measured amplifier gain as a function of frequency.

5.8 D2. 1-dB Compression Point

The input power of the amplifier was gradually increased while measuring the corresponding output power. The gain was calculated from

$$G = P_{\text{out}} - P_{\text{in}}. \tag{27}$$

Table 6: Amplifier compression measurements.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
-20	2.68	22.68
-19	3.67	22.67
-18	4.64	22.64
-17	5.64	22.64
-16	6.62	22.62
-15	7.62	22.62
-14	8.60	22.60
-13	9.62	22.62
-12	10.58	22.58
-11	11.54	22.54
-10	12.52	22.52
-9	13.53	22.53
-8	14.42	22.42
-7	15.20	22.20
-6	16.23	22.23
-5.5	16.28	21.78
-5.4	16.32	21.72
-5.3	16.47	21.77
-5	16.70	21.70
-4	17.20	21.20
-3	17.40	20.40
-2	17.54	19.54
-1	17.61	18.61
0	16.28	16.28

The small-signal gain was obtained by averaging the gain at the five lowest input powers:

$$G_0 = \frac{22.68 + 22.67 + 22.64 + 22.64 + 22.62}{5} \approx 22.65\,\mathrm{dB}.$$

At the 1-dB compression point,

$$G_{1\mathrm{dB}} = G_0 - 1 = 21.65\,\mathrm{dB}.$$

From the measured data,

$$G(-5\,\mathrm{dBm}) = 21.70\,\mathrm{dB}$$

and

$$G(-4\,\mathrm{dBm}) = 21.20\,\mathrm{dB}.$$

Linear interpolation gives

$$P_{1\mathrm{dB,in}} = -5 + \frac{21.65 - 21.70}{21.20 - 21.70}(1) \approx -4.9\,\mathrm{dBm}.$$

The corresponding output power is approximately

$$P_{1\mathrm{dB,out}} \approx 16.8\,\mathrm{dBm}.$$

Thus,

$$\boxed{P_{1\mathrm{dB,in}} \approx -4.9\,\mathrm{dBm}}$$

and

$$\boxed{P_{1\mathrm{dB,out}} \approx 16.8\,\mathrm{dBm}}.$$

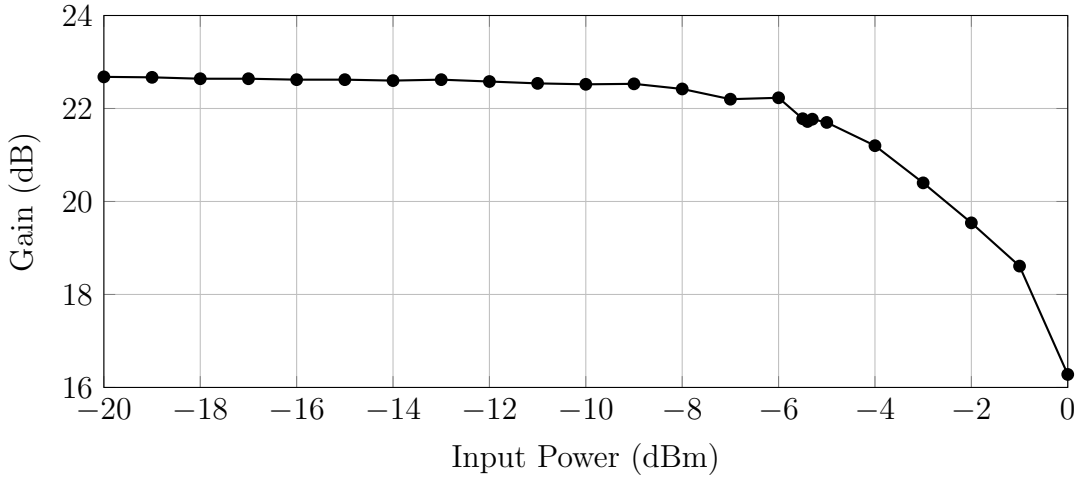


Figure 4: Amplifier gain as a function of input power showing gain compression.

5.9 D3. Third-Order Intercept Point

Two RF tones were applied at

$$f_1 = 10\,\mathrm{MHz}, \qquad f_2 = 13\,\mathrm{MHz}.$$

The third-order intermodulation products occur at

$$2f_1 - f_2 = 7\,\mathrm{MHz}$$

and

$$2f_2 - f_1 = 16\,\mathrm{MHz}.$$

The input power per tone was varied and the fundamental output power and third-order intermodulation (IM3) power were measured.

Table 7: Two-tone intermodulation measurements.

P_{in} per tone (dBm)	P_{fund} (dBm)	P_{IM3} (dBm)
-20	0.74	-61.20
-18	2.76	-57.60
-16	4.70	-51.20
-14	6.67	-45.09
-12	8.62	-38.25
-10	10.47	-25.07

The fundamental and IM3 powers were fitted as functions of input power. For the fundamental, the fitted relation was

$$P_{\text{fund}} = P_{\text{in}} + 20.66,$$

while for the third-order product,

$$P_{\text{IM3}} = 3P_{\text{in}} - 1.40.$$

The intersection of the two extrapolated lines gives the input-referred third-order intercept point:

$$\boxed{P_{\text{in,TOI}} \approx 11.0\,\text{dBm}}$$

per tone.

Using the small-signal gain of approximately 20.66 dB, the corresponding output-referred TOI is

$$\boxed{P_{\text{out,TOI}} \approx 31.7\,\text{dBm}}.$$

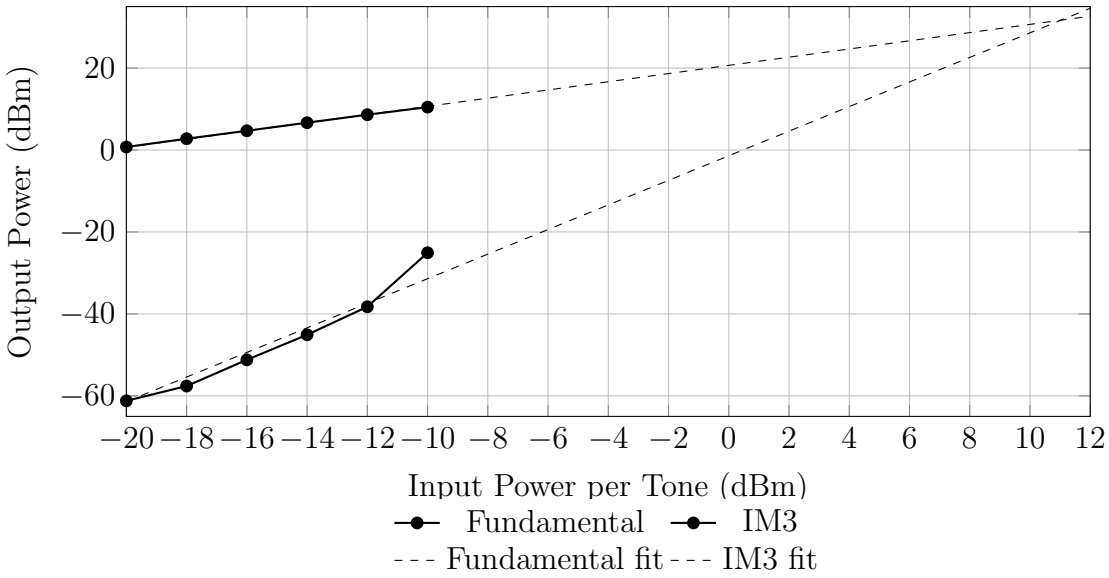


Figure 5: Fundamental and third-order intermodulation output powers as functions of input power. The extrapolated lines intersect at the third-order intercept point.

6 Results

The main experimental results are summarized below:

- Filter cutoff frequency: $\boxed{f_c \approx 548\,\text{MHz}}$; return loss: $\boxed{36.5\,\text{dB}}$.
- Stub notch frequency: $\boxed{55\,\text{MHz}}$, with maximum attenuation of $\boxed{28.5\,\text{dB}}$.
- Mixer average conversion loss: $\boxed{11.4\,\text{dB}}$.
- Splitter average losses: $\boxed{3.55\,\text{dB}}$ and $\boxed{3.52\,\text{dB}}$.
- Combiner output power: $\boxed{3.05\,\text{dBm}}$.
- Amplifier gain: approximately $\boxed{22.2\text{–}22.7\,\text{dB}}$ over 100–200 MHz.
- 1-dB compression point: $\boxed{P_{1\text{dB,in}} \approx -4.9\,\text{dBm}}$.
- Output-referred TOI: $\boxed{P_{\text{out,TOI}} \approx 31.7\,\text{dBm}}$.

7 Conclusion

The major RF components were successfully characterized experimentally. The filter, stub, mixer, splitter/combiner, and amplifier showed the expected RF behaviour. Important parameters such as cutoff frequency, conversion loss, insertion loss, gain, 1-dB compression point, and third-order intercept point were determined from the measurements.

8 References

Department Laboratory Manual, “RF Components: Characterization and Measurement,” SPA617 Experimental Manual.

Use of AI

AI tools were used only as an assistive tool for language refinement, LaTeX formatting, and report organization. All experimental data, calculations, graphs, and reported results were obtained from the laboratory work that I performed.